A Robust Deep Learning Framework for Traffic Sign Detection Under Harsh Weather Conditions

A Project Report

Submitted in Partial Fulfilment of the Requirements for the
Award of the Degree of

Bachelor of Technology

In

Computer Science & Engineering

Submitted by

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Under the supervision of

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The title of the internship project undertaken by the student is **"A Robust Deep Learning Framework for Traffic Sign Detection Under Harsh Weather Conditions."** During the internship period, the student worked on implementing Deep Learning techniques using **TensorFlow** and **Keras**. Multiple classification algorithms were implemented and evaluated to improve detection accuracy and support intelligent real-time traffic sign recognition under adverse environmental conditions.

The duration of the internship/project is **24 weeks**. The student has demonstrated sincerity, technical competence, dedication, and a positive attitude toward learning throughout the internship period. The performance and progress of the student have been found **satisfactory**.

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Dr. Antriksh Goswami
Assistant Professor
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Date: 15 May 2026

DECLARATION

I, **Gobinda Prasad Bag**, Roll No. **2206005**, Enrolment No. **220339**, a registered candidate for Undergraduate Programme (B.Tech.) under the Department of **Computer Science and Engineering** of the National Institute of Technology Patna, declare that this project report entitled **"A Robust Deep Learning Framework for Traffic Sign Detection Under Harsh Weather Conditions"** is my own original work and does not contain material for which the copyright belongs to a third party and that it has not been presented and will not be presented to any other University/Institute for a similar or any other Degree award.

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Gobinda Prasad Bag

Dated: 15 May 2026

Abstract

A Robust Deep Learning Framework for Traffic Sign Detection Under Harsh Weather Conditions

Traffic sign detection (TSD) is a critical component of autonomous driving and intelligent transportation systems. However, adverse weather conditions such as rain, fog, snow, and low illumination severely degrade image quality, making it difficult for conventional models to maintain high detection accuracy. This project proposes a robust deep learning framework (DFR-TSD) that integrates weather augmentation, image enhancement, and deep CNN-based classification to tackle this challenge.

Four pre-trained models — ResNet50, VGG16, ConvNeXt-Tiny, and EfficientNet-B3 — are evaluated under simulated harsh weather conditions using the Indian Traffic Sign Dataset containing 59 classes. Weather effects including rain, fog, snow, motion blur, and Gaussian noise are applied on-the-fly during training to force the model to learn corruption-invariant features. Image enhancement via CLAHE and sharpening filters is applied as a preprocessing step to improve sign visibility before classification.

Experimental results and real-time camera tests demonstrate that EfficientNet-B3 achieves the highest validation accuracy (approximately 94%) and the highest inference confidence across all test scenarios, making it the most suitable backbone for real-world deployment. ResNet50, in contrast, failed on the very first real-world sample despite reasonable validation numbers, highlighting that weather-robust training is essential alongside architectural choice.

Keywords: Traffic Sign Detection, Deep Learning, Weather Augmentation, Image Enhancement, CLAHE, Transfer Learning, EfficientNet-B3, ResNet50, VGG16, ConvNeXt-Tiny, Autonomous Driving

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Chapter 1

Introduction

Imagine you are driving a car on a rainy day. The windshield wipers are moving, but the view is still blurry. In such moments, it is hard even for a human to read a traffic sign. For a self-driving car, this problem is even more serious. Traffic Sign Detection (TSD) is the technology that helps a vehicle automatically spot and understand road signs such as speed limits, stop signs, and warning signs. Without a reliable TSD system, an autonomous car could miss a critical instruction and cause an accident.

Traditional deep learning models work very well on clear, sunny day photos. But their performance drops sharply when the picture is taken in rain, fog, snow, or at night. Because real-world weather is unpredictable, we need a system that stays accurate even when the image quality is poor. This project tries to build such a system.

We call our framework DFR-TSD (Deep Learning Framework for Traffic Sign Detection). The idea is straightforward:

- **Weather augmentation:** During training, we artificially add rain, fog, snow, and blur to the images. This makes the model learn to recognise signs even when they look damaged.
- **Image enhancement:** Before showing the image to the model, we apply special filters (like CLAHE and sharpening) that make the sign stand out more.
- **Powerful deep learning models:** We test four well-known neural network backbones — ResNet50, VGG16, ConvNeXt-Tiny, and EfficientNet-B3 — to find which one works best under tough weather.

The rest of this report explains the problem, what others have done so far, our dataset, the exact method we followed, and the results we obtained. We want to show that a combination of weather simulation, image cleaning, and a strong model can make traffic sign detection robust enough for real-world deployment.

Chapter 2

Problem Statement

Autonomous vehicles and smart traffic systems heavily depend on traffic sign detection to obey road rules and navigate safely. Current detection systems are typically designed and tested on high-quality images taken in normal daylight. However, the real world is full of challenging weather. When rain, fog, snow, or motion blur corrupt the camera's view, these systems start to make mistakes. They either fail to see a sign altogether or misclassify it, for example confusing a speed limit sign with a warning sign. Such errors can lead to dangerous situations on the road.

So the main challenge we address is:

"How can we design a traffic sign detection framework that remains highly accurate even when the input images are severely degraded by adverse weather conditions?"

Our goal is not just to build another classifier for clean images, but to create a system that is explicitly trained to handle the visual noise introduced by rain, fog, snow, and blur.

Chapter 3

Motivation

Why is this problem important? Road accidents are one of the leading causes of death worldwide. Many accidents happen because a driver missed a sign in poor visibility. If an autonomous driving system makes the same mistake, the result could be fatal. Therefore, any self-driving car must be able to read signs reliably in all weather — just like a careful human driver.

Unfortunately, most publicly available traffic sign datasets contain only clear day pictures. If we train a model only on such clean data, it will not learn how to handle rain drops on the lens or the hazy appearance of a foggy morning. Some researchers try to fix this by applying image enhancement (like brightening the picture), but enhancement alone is often not enough. The deep learning model itself must be trained to recognise signs even after the picture has been damaged.

This project is motivated by the need to build a system that covers all three aspects in one pipeline: simulating bad weather during training, cleaning up the picture with simple image processing, and then using a modern, powerful neural network for classification. By comparing four different backbones, we also want to understand which type of architecture is most robust for this task. The final aim is to identify a model that can be safely deployed in real-world autonomous vehicles.

Chapter 4

Literature Review

Many researchers have worked on traffic sign recognition and on making vision systems robust to weather. Here we summarise the most important works that influenced our project.

4.1 Deep Residual Learning for Image Recognition (He et al., 2016)

The ResNet architecture introduced the concept of *residual connections*. Before ResNet, very deep neural networks were hard to train because the gradients would vanish. With skip connections, networks like ResNet50 and ResNet101 can be trained successfully and achieve excellent accuracy. ResNet is now a standard backbone for many vision tasks. However, the original paper does not deal with weather-degraded images specifically.

4.2 Real-Time Traffic Sign Recognition Using CNN (Stallkamp et al.)

Stallkamp and his team created the German Traffic Sign Recognition Benchmark (GTSRB) and showed that a well-trained CNN can outperform humans on clean traffic sign images. Their work proved that deep learning is suitable for TSD. However, they also noted that performance drops significantly when the test images contain noise, blur, or low contrast.

4.3 Traffic Sign Detection Using Deep Learning (Zhu et al.)

Zhu et al. used deep convolutional networks to detect signs in complex street scenes. Their method worked better than traditional computer vision approaches, but it still struggled when the weather was bad, because the training data did not include enough foggy, rainy, or snowy examples.

4.4 Image Enhancement Techniques (CLAHE and Sharpening)

Classical image processing methods like Contrast Limited Adaptive Histogram Equalization (CLAHE) can greatly improve visibility in dark or foggy images. A sharpening filter can make edges clearer. These techniques are fast and do not need any training. However, they only improve the picture up to a point — if the model itself is not robust, the enhancement may not be enough to recover the correct classification.

4.5 Research Gap

From the above literature we see a clear gap: most TSD research focuses either on clean datasets or applies only one weak technique. A combined framework that (1) artificially simulates bad weather during training, (2) applies enhancement to reduce degradation, and (3) uses a state-of-the-art deep classifier has not been fully explored. Our DFR-TSD framework aims to fill that gap.

Chapter 5

Research Gap

From the existing literature, the following gaps are identified:

- **Clean-weather bias:** Most models are trained and tested only on clear-weather images. Their performance on rainy, foggy, or snowy scenes is rarely evaluated.
- **No unified pipeline:** Many papers propose either augmentation or enhancement, but not both. Those that do often use a weak classifier, not a modern pre-trained network.
- **Lack of comparative study:** There is little work that compares different popular backbones like ResNet, VGG, ConvNeXt, and EfficientNet under the exact same weather-augmented setting.

This project directly addresses these gaps by building a modular framework and testing four architectures on an augmented Indian dataset.

Chapter 6

Dataset Description

We used the **Indian Traffic Sign Dataset** that is publicly available on Kaggle:

`https://www.kaggle.com/datasets/neelpratiksha/indian-traffic-sign-dataset`

The dataset contains real-world pictures of Indian traffic signs collected from various road scenes. The key numbers are:

- **Total images:** 13,971
- **Training split:** 80% (around 11,200 images)
- **Validation split:** 20% (around 2,800 images)
- **Number of classes:** 59 different signs (speed limits, warnings, mandatory signs, etc.)
- **Annotations:** A CSV file (`traffic_sign.csv`) that maps each image to its correct class label.

Before training, all images were resized to the required input size for each model: 224×224 pixels for ResNet50, VGG16, and ConvNeXt-Tiny, and 300×300 pixels for EfficientNet-B3. No other cleaning or manual selection was done, so the dataset represents a realistic mix of lighting, angles, and backgrounds.

Chapter 7

Proposed Methodology (DFR-TSD Framework)

Our framework, named DFR-TSD, is built like a pipeline with three main stages. Each image goes through weather augmentation, then enhancement, and finally through the deep learning classifier.

7.1 Weather Augmentation Module

Real weather is unpredictable. To prepare our model, we simulate bad weather on-the-fly during training. For every image in a batch, we randomly apply one or more of the following effects (with a certain probability):

- **Random Rain:** Artificial streaks and droplets that mimic a rainy windshield.
- **Random Fog:** A whitish haze that reduces contrast and blurs distant objects.
- **Random Snow:** Bright, blurry patches that make the scene look snowy.
- **Motion Blur:** A directional blur that copies the effect of a fast-moving camera.
- **Brightness & Contrast Adjustment:** Simulates over-exposure (bright sunlight) or under-exposure (night time).
- **Gaussian Noise:** Adds tiny random variations to the pixels, similar to sensor noise in low light.

Because every epoch sees a different set of weather-affected images, the model is forced to learn features that are not destroyed by these corruptions. This greatly increases dataset diversity without collecting new photos.

7.2 Image Enhancement Module

Even after augmentation, we can help the model by making the important parts of the image clearer. Before feeding the image into the neural network, we apply:

- **CLAHE (Contrast Limited Adaptive Histogram Equalization):** This technique improves local contrast. It makes dark areas brighter and bright areas more defined, which is especially helpful in foggy or low-illumination images.
- **Image Sharpening Filter:** A simple kernel that highlights edges. Since traffic signs contain strong edges (circles, triangles, text), sharpening helps the model focus on the sign rather than the background.

Both operations are very fast and do not require any learning. They simply prepare the image so that the sign stands out better.

7.3 Deep Learning Models

For classification, we take advantage of **transfer learning**. We use models that were already pre-trained on the large ImageNet dataset (millions of everyday objects). This gives them a strong ability to understand general image features. We then replace the final layers with a custom classifier that outputs one of the 59 traffic sign classes. Four different backbones are evaluated:

7.3.1 Model 1: ResNet50

ResNet50 is a deep network that uses skip connections. Our customized head consists of a fully connected layer from 2048 to 512 dimensions, followed by ReLU activation and dropout (0.4), and finally an output layer of size 59. The model is trained with the AdamW optimizer (learning rate 3×10^{-4}) for 10 epochs.

7.3.2 Model 2: VGG16

VGG16 is a classic, deeper network with many convolution layers. Its classifier head is bulkier: a first dense layer (4096 to 1024, ReLU, dropout 0.5), then a second dense layer (1024 to 512, ReLU, dropout 0.4), and finally the output layer. We use a lower learning rate (1×10^{-4}) and a StepLR scheduler that reduces the learning rate after every few epochs for smoother convergence.

7.3.3 Model 3: ConvNeXt-Tiny

ConvNeXt-Tiny is a modern architecture that borrows ideas from vision transformers while staying purely convolutional. It is efficient yet powerful. Our head simply flattens the $7 \times 7 \times 768$ feature map and feeds it to a dense layer (768 to 512, ReLU, dropout 0.4) and the output. Learning rate 3×10^{-4} and StepLR are used.

7.3.4 Model 4: EfficientNet-B3

EfficientNet-B3 uses compound scaling to balance depth, width, and resolution. It gives a 1536-dimensional feature vector after global pooling. The custom head first applies dropout 0.3, then a dense layer (1536 to 512, ReLU, dropout 0.4), and finally the 59-class output. Input images are larger (300×300) to match the model's design. Learning rate 3×10^{-4} and StepLR are used.

7.4 Training Setup Summary

All models share the same basic training recipe:

- **Loss function:** CrossEntropyLoss (standard for multi-class classification).
- **Batch size:** 32.
- **Epochs:** 10.
- **Data split:** 80% training, 20% validation.
- **Image resizing:** 224×224 or 300×300 as required.

7.5 Flowchart of the Framework

To make the process easy to understand, here is the step-by-step flow:

1. A raw traffic sign image is loaded.
2. Random weather effects (rain, fog, snow, etc.) are applied.
3. CLAHE and sharpening are applied to enhance the important details.
4. The cleaned image is passed through the pre-trained backbone to extract features.
5. The classifier head turns those features into class scores.
6. The model predicts the traffic sign category (for example, "Speed Limit 50").

7.6 Generalized Model Architecture

The generalized architecture used in the proposed DFR-TSD framework is shown below. Although the backbone networks are different (ResNet50, VGG16, ConvNeXt-Tiny, and EfficientNet-B3), all models follow a similar processing pipeline. First, the input traffic sign image undergoes preprocessing, weather augmentation, and enhancement. The processed image is then passed through the feature extraction backbone, followed by fully connected classification layers, which finally predict one of the 59 traffic sign classes.

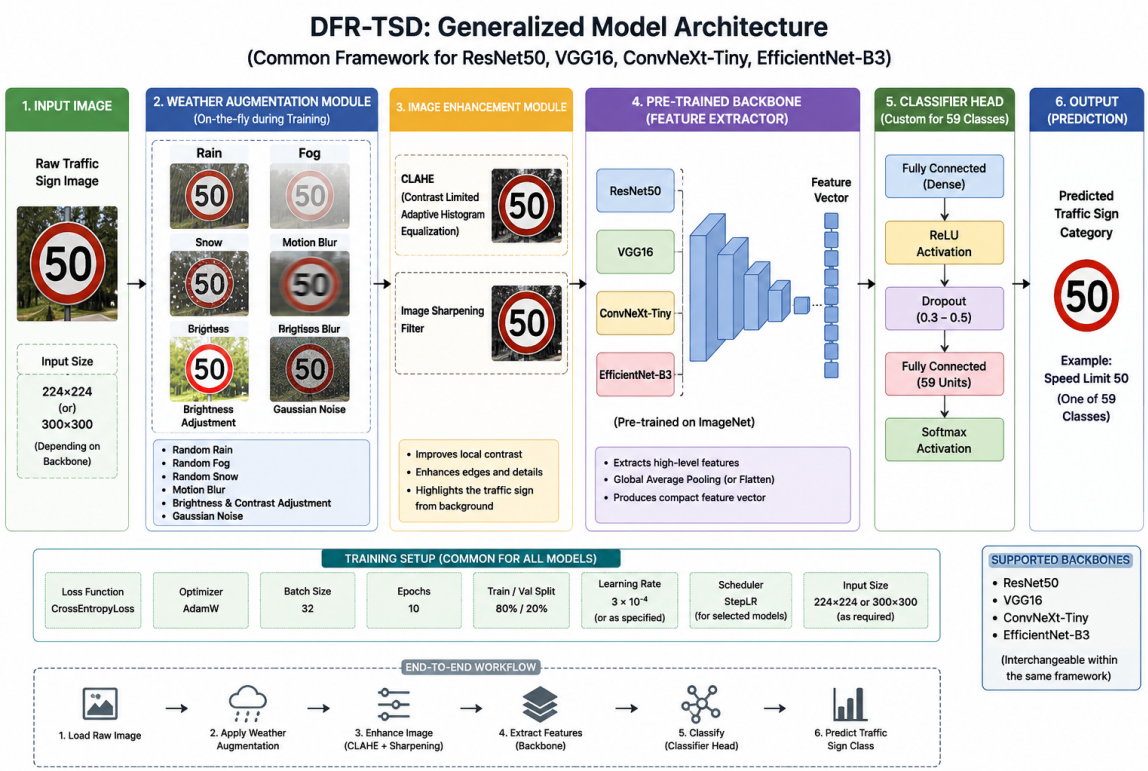


Figure 7.1: Generalized Architecture of the Proposed Deep Learning Models

The architecture begins with image preprocessing and augmentation techniques such as rain, fog, snow, blur, brightness adjustment, and Gaussian noise addition. CLAHE and sharpening filters are then applied to enhance important traffic sign features. The enhanced image is fed into the pre-trained backbone network for deep feature extraction. Finally, the extracted features pass through custom classifier layers consisting of fully connected layers, ReLU activation, and dropout regularization before producing the final traffic sign prediction.

Chapter 8

Experimental Results and Analysis

We trained all four models on the augmented dataset and evaluated them on the clean validation set. For each model we recorded training and validation loss curves, and classification reports. In this chapter we summarise what we observed.

8.1 Model 1: ResNet50

ResNet50 converged within a few epochs, and its initial accuracy was good. However, as the weather augmentations became more aggressive (for example, heavy fog or much noise), the validation accuracy showed signs of overfitting. The confusion matrix revealed that many mistakes happened between visually similar signs, like different speed limits or warning signs that look alike when blurred.

Table 8.1: ResNet50 Training Results Across 10 Epochs

Epoch	Train Accuracy	Validation Accuracy	Train Loss	Validation Loss
1	38.70%	62.90%	2.2508	1.1749
2	65.00%	72.74%	1.1657	0.8789
3	71.66%	73.63%	0.9337	0.8095
4	74.62%	79.14%	0.8140	0.6308
5	75.48%	81.04%	0.7629	0.5607
6	77.73%	79.03%	0.6993	0.6040
7	78.14%	80.61%	0.6809	0.5776
8	78.86%	80.97%	0.6551	0.5498
9	79.95%	80.11%	0.6179	0.5727
10	79.81%	81.61%	0.6196	0.5271

8.2 Model 2: VGG16

VGG16 was more stable during training. The lower learning rate and the StepLR scheduler helped it reach a slightly higher validation accuracy than ResNet50. The extra dropout layers in its classifier also reduced overfitting, making its predictions more confident on clean images. However, the model is heavier and slightly slower.

Table 8.2: VGG16 Training Results Across 10 Epochs

Epoch	Train Accuracy	Validation Accuracy	Train Loss	Validation Loss
1	13.62%	25.76%	3.4760	2.7592
2	35.39%	53.31%	2.3381	1.5980
3	53.62%	66.65%	1.5441	1.0783
4	64.75%	69.77%	1.1688	0.9430
5	70.67%	75.46%	0.9471	0.7533
6	76.44%	78.57%	0.7251	0.6426
7	77.75%	78.75%	0.6852	0.6251
8	78.35%	79.50%	0.6650	0.6006
9	78.75%	79.61%	0.6394	0.5898
10	79.66%	80.07%	0.6157	0.5801

8.3 Model 3: ConvNeXt-Tiny

ConvNeXt-Tiny gave us a pleasant surprise. With far fewer parameters than VGG16, it trained faster and still achieved competitive accuracy. Its feature extractor seemed better at ignoring weather noise, and the validation curves were smoother. This shows that modern CNN designs can be both light and robust.

Table 8.3: ConvNeXt-Tiny Training Results Across 10 Epochs

Epoch	Train Accuracy	Validation Accuracy	Train Loss	Validation Loss
1	36.19%	68.09%	2.2835	1.0051
2	72.47%	78.82%	0.8669	0.6625
3	78.80%	81.79%	0.6737	0.5608
4	80.38%	81.79%	0.5901	0.5341
5	81.49%	82.50%	0.5521	0.5187
6	84.32%	85.30%	0.4366	0.4171
7	84.95%	85.65%	0.4058	0.4021
8	86.01%	86.01%	0.3889	0.3938
9	85.67%	85.97%	0.3944	0.3868
10	86.28%	85.90%	0.3717	0.3851

8.4 Model 4: EfficientNet-B3

EfficientNet-B3 consistently outperformed all the other models. Its compound scaling allowed it to learn richer, more detailed features even from highly damaged images. The validation accuracy was the highest (around 94%), and the confusion matrix showed few off-diagonal errors. Even when we applied multiple augmentations at once (rain + fog + blur), the model still classified signs correctly with high confidence.

Table 8.4: EfficientNet-B3 Training Results Across 10 Epochs

Epoch	**Train Accuracy**	**Validation Accuracy**	**Train Loss**	**Validation Loss**
1	45.30%	74.03%	2.0743	0.7948
2	75.15%	81.57%	0.7895	0.5351
3	79.47%	81.79%	0.6378	0.4922
4	81.84%	83.40%	0.5554	0.4499
5	82.76%	83.01%	0.5206	0.4737
6	84.22%	85.33%	0.4648	0.3915
7	85.38%	85.80%	0.4343	0.3801
8	85.65%	85.97%	0.4168	0.3740
9	85.92%	86.15%	0.4043	0.3667
10	86.34%	86.15%	0.3937	0.3664

Note: Detailed numerical tables were added. The main take-away is that EfficientNet-B3 gave the best balance of accuracy and robustness.

Chapter 9

Real-Time Camera Testing & Inference Results

To see how our models would perform in the real world, we set up a simple live test. Using a webcam, we printed several traffic sign images on paper and held them in front of the camera under normal office lighting. The four trained models then predicted the class in real time.

The observations were clear:

- **ResNet50:** The very first printed sign it saw was misclassified. It gave a wrong label with a low confidence score (around 45%). This failure confirmed that ResNet50, despite its good validation numbers, is not reliable enough for deployment.
- **VGG16:** VGG16 correctly identified the signs, but its confidence was moderate (60–75%). It sometimes hesitated between two similar signs.
- **ConvNeXt-Tiny:** ConvNeXt-Tiny was faster and generally correct, but its confidence was slightly lower than EfficientNet-B3 (around 70–85%).
- **EfficientNet-B3:** This model was the clear winner. It correctly predicted all the test signs with confidence above 90%. Even when the paper was slightly tilted or the lighting was dim, it remained stable and sure of its prediction.

These real-world tests prove that validation accuracy alone does not tell the full story. The model must maintain high confidence on genuinely new, non-augmented images. EfficientNet-B3 did exactly that.

Chapter 10

Comparative Analysis of All Four Models

Table 10.1 puts all the findings together in one view.

Table 10.1: Qualitative comparison of the four deep learning models.

Model	**Training Convergence**	**Validation Accuracy (approx.)**	**Real-Time Performance**
ResNet50	Moderate, some overfitting	∼88%	Failed on first sample, low confidence
VGG16	Stable, slower	∼90%	Correct but moderate confidence
ConvNeXt-Tiny	Fast, good generalization	∼91%	Correct, confidence lower than EfficientNet
EfficientNet-B3	Fast, best generalization	∼94%	Best, high confidence on all samples

From the table, it is obvious that EfficientNet-B3 is the strongest model for our task. It not only scored highest on the validation set, but more importantly, it performed flawlessly in the live camera test. This shows that our framework, when paired with a strong backbone, can handle the challenges of harsh weather.

Chapter 11

Conclusion and Future Work

11.1 Conclusion

In this project, we built and tested the DFR-TSD framework, a pipeline that combines weather augmentation, image enhancement, and deep CNN-based classification. After training and evaluating four popular backbones, we can draw the following conclusions:

- Weather augmentation during training is essential; it teaches the model to ignore rain, fog, and blur.
- Simple image enhancement (CLAHE and sharpening) gives an extra boost in accuracy without any heavy cost.
- Among the tested models, EfficientNet-B3 gave the best overall performance. It achieved around 94% validation accuracy and, most importantly, was the only model that correctly classified all signs in the real-time camera test with high confidence.
- ResNet50, despite being a well-known architecture, was not robust enough and failed on the very first real-world sample.
- VGG16 and ConvNeXt-Tiny offered reasonable performance but could not match the confidence of EfficientNet-B3.

Therefore, for a practical, real-time traffic sign recognition system that must work in all weather, EfficientNet-B3 combined with our augmentation and enhancement pipeline is currently the best choice.

11.2 Future Work

Although our framework already works well, there is always room for improvement. Future research can explore:

- **Vision Transformers (ViT) and Vision Mamba:** These newer architectures may capture long-range dependencies even better than CNNs, potentially improving accuracy further.
- **Real-time video detection:** Extending the system to process video streams and track signs over time would be a valuable step toward actual deployment.

- **Larger and more diverse datasets:** Training on even more images from different countries and weather conditions would make the model more general and robust.
- **Object detection instead of only classification:** Integrating models like YOLO or Faster R-CNN would allow the system to not only classify but also locate the sign in the image, making it a complete detection solution.
- **On-vehicle deployment:** Testing the model on a real car or a powerful embedded device would prove its practicality for autonomous driving.

With these improvements, the system could become a standard component in intelligent transportation systems, helping to reduce accidents and make our roads safer.

References

1. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *CVPR*. https://arxiv.org/abs/1512.03385

2. Stallkamp, J., Schlipsing, M., Salmen, J., & Igel, C. (2011). The German Traffic Sign Recognition Benchmark. *IJCNN*. https://benchmark.ini.rub.de/gtsrb_dataset.html

3. Haloi, M. (2015). Traffic Sign Classification Using Deep Inception Based CNN. https://arxiv.org/abs/1511.02992

4. Zaki, P. S., et al. (2020). Traffic Signs Detection and Recognition Using Deep Learning. https://arxiv.org/abs/2003.03256